NO. 71.5 · SINGAPORE

A SPECIAL EDITION OF

MERIDIAN

Shortwave listening — an ultra deep dive

A dial, tuned between stations — representative image, not a specific receiver or broadcast. Source: Pexels.

THE CEILING

A US$16,000 flagship, a Cold War "boat anchor" that still out-sells new gear on eBay, and the antenna maker whose closure left a hole nobody has fully filled

THE SCEPTIC

Wikipedia's own entry says there are few practical reasons left to buy one — the hobby's hardest cross-examination, taken seriously

THE LONG READ

A hundred antennas playing keepaway across three countries, a jammer that has never once identified itself, and the live radio war Western coverage keeps filing as a footnote

Reader-commissioned special · 13 August 2026

THIS SPECIAL

Shortwave Radio — Contents

FROM THE DESK

"This... is London."

— Edward R. Murrow, opening his wartime CBS shortwave broadcasts from London, 1940[1]

Murrow's signature pause named a whole medium's promise — a live voice, carried nowhere but through the air, from a city under siege to a living room an ocean away. Eighty-six years later, the transmitting station built to carry his network's successor still bears his name, and its lights went dark and flickered back on inside the twelve months this book covers.[2]

SHORTWAVE LISTENING — AT A GLANCE

METRIC	FIGURE
Global shortwave radios market, 2025	~US$1.38B
Serious day-one starter kit	~US$242
The DXer's recommended SDR	Airspy HF+ Discovery, US$169
The flagship tabletop	Icom IC-R8600, ~US$2,750
Sets estimated worldwide	~600 million
Solar Cycle 25 peak	October 2024 — still elevated

Figures as researched for this edition, Aug 2026; see the sections named for full sourcing.[3]

WHY THIS SPECIAL EXISTS

This edition was commissioned directly, off the Rabbit Hole's three-page treatment in MERIDIAN No. 71 (p15–17): "The Band Everyone Left, and the Hobby That Wouldn't Die With It." That piece established the paradox (a shrinking band, a growing hobby), the SDR-vs-classic-portable split, the antenna-first thesis, and the Singapore basics. This book assumes the reader has read it and goes everywhere that piece did not have three pages to go: the full propaganda-and-jamming history, the complete gear ladder through a genuine restoration-and-boutique-maker tier, the last twelve months named and dated, a harder cross-examination, and a closing long read into a live radio war Western coverage has mostly filed as a press-freedom footnote rather than the technically fascinating story it is.[4]

SOURCES 1 Edward R. Murrow's wartime London broadcasts — widely documented opening formula, CBS Radio, 1940; Cold War Radio Museum; Smithsonian Magazine on Allied shortwave propaganda operations. 2 Edward R. Murrow Transmitting Station, Greenville, NC — International Broadcasting Bureau Greenville Transmitting Station, Wikipedia; see p5–6 for the 2025–26 shutdown-and-partial-return sourcing. 3 See the named sections for individual sourcing; market-size figures are directional industry estimates, flagged as such on p17. 4 MERIDIAN No. 71 (daily), "The Rabbit Hole · The Band Everyone Left."

A NEW COMMISSION

The Invitation

The daily's version of this hobby fit on three pages because three pages is what a Tuesday allows. This one exists because a reader asked for the rest — the history, the full gear ladder, the twelve months just gone, and the one story in this world that Western coverage still hasn't found.

this story 👍 👎

Start with the number that should not make sense. The global market for shortwave radios was worth roughly US$1.38 billion in 2025, and the adjacent transmitter-equipment market — the industrial side, not the hobby side — is tracked at close to a billion dollars and growing at better than five per cent a year through the early 2030s.[1] Meanwhile the roll call of institutions that gave the hobby something to listen to keeps getting shorter: the BBC World Service off shortwave to North America in 2001; Radio Australia off the band entirely on 31 January 2017; Deutsche Welle's linear shortwave wound down through 2011; Radio Netherlands gone by 2014; and, most dramatically, the Voice of America's flagship US transmitters silenced by executive order in March 2025.[2] A market growing while its supply of content shrinks is not how markets are supposed to behave. The resolution is that shortwave listening was never really a content business — it is an infrastructure hobby that happens to have been subsidised, for most of a century, by the free content that Cold War propaganda budgets and colonial-era broadcasting networks provided. Take the free content away and the infrastructure hobby is still there, now paying its own way.

The daily's own three pages already made this case in miniature: the entry price collapsed from a roughly US$200 receiver a decade ago to a roughly US$30 SDR dongle today, and three forces — the price collapse, a documented surge in prepper and grid-down buying, and an open-ended supply of new things to *decode* rather than merely hear — are pulling in new listeners faster than the broadcasters are leaving.[3] This book takes that as settled and asks the harder questions the format didn't have room for. Is the growth real hobbyist growth, or is it mostly disaster-preparedness commerce that never leads to an actual logbook entry — and does that distinction even matter? What, precisely, is left worth chasing once the marquee broadcasters are gone? And where, if anywhere, is someone fighting to stay on this band rather than quietly leaving it?

US$1.38B

ESTIMATED GLOBAL SHORTWAVE-RADIOS MARKET, 2025 — A MARKET-RESEARCH FIGURE, DIRECTIONAL NOT EXACT, BLENDING HOBBYIST AND EMERGENCY-PREPAREDNESS DEMAND[1]

> ***A market growing while its supply of content shrinks is not how markets are supposed to behave.***

THE TWO BOOKS THIS SPECIAL IS NOT

This is not a how-to for a first-time buyer — the daily's own piece and a hundred YouTube channels named on p7 already do that well. It is also not a nostalgia piece for a dying medium; the Sceptic (p19–20) takes the decline case seriously enough that a nostalgia frame would be dishonest. This book is the buyer's guide, the community map and the honest economics for a reader who has already decided the subject interests him and wants to know what it actually costs, who actually runs it, and where it is still, provably, alive.[4]

SOURCES 1 Global shortwave radios market ~US$1.38B (2025) — Dataintelo, "Shortwave Radios Market Research Report 2034"; shortwave transmitter equipment market ~US$935M (2025) rising to ~US$1.35B by 2032 at 5.36% CAGR — 360iResearch, "Shortwave Radio Transmitters Market"; both flagged as directional industry estimates. 2 Broadcaster exodus timeline — BBC World Service (Radio World; Al Jazeera); Radio Australia 31 Jan 2017 (Radio World); Deutsche Welle 2011 (Radiomuseum.org forum); Radio Netherlands 2012–14 (Wikipedia, "Radio Netherlands Worldwide"); Voice of America March 2025 (NPR; see p5 for full sourcing). 3 MERIDIAN No. 71 (daily), "The Rabbit Hole · The Band Everyone Left," p15. 4 Editorial framing, this edition.

This book runs outward the way MERIDIAN's hobby specials always do. The Long Wave (p5) is the history the daily had no room for at all — the 1930s propaganda war, the Cold War jamming war, and the broadcaster retreat still unfolding as this book was written. The Community (p7) maps the hub blog, the forum split, the numbers-station monitoring groups, and the reference annual quietly retiring its print edition after eighty years. The Ladder (p9) and The Ceiling (p11) are the gear, argued the way this reader's other hobbies have trained him to expect: the quality entry point as the real entry point, running all the way to a genuine restoration-and-boutique-antenna tier this hobby has instead of a commissioned-maker one. The Cutting Edge (p13) is the last twelve months, named, dated, and honestly graded for whether any of it is real progress or just necessary maintenance. The Antenna (p15) is the craft itself — propagation, modes, logging — plus a glossary. The Economics (p17) is the real cost at every level. The Sceptic (p19) is the hard cross-examination, including two arguments the daily's three pages never got to. The Little Red Dial (p21) is the honest Singapore angle. The First Year (p23) is a real starting path. And the Long Read (p25) follows the hobby's oldest use — carrying a signal past a government that doesn't want it heard — into the one place that fight is still genuinely, currently live.

A word on continuity, because it matters for how to read this book. The daily's piece already established the SDR-versus-classic-portable split, the antenna-first thesis for an urban listener, and the Singapore basics — that receiving needs no licence, that the local amateur exam is suspended, and that a dense HDB block is close to a worst-case radio-frequency noise environment.[1] This book does not re-argue those points from scratch; it takes them as read and goes further, with a Singapore section that tried to verify the licensing claim harder than a Tuesday allows — and reports honestly on where that verification ran out of road.[2]

A receiver that needs no grid or internet has provable value the one week it matters — that case is orthogonal to whether the sport of DXing is as good as it used to be.

THE CROSS-EXAMINATION, EARLY

The obvious objection: everything on shortwave streams online now, so why own a radio at all? The Sceptic, on p19–20, takes that seriously — including Wikipedia's own entry on the hobby, which concedes there are few practical reasons left to buy a shortwave set, and a harder look at whether prepper commerce is inflating this hobby's apparent health relative to its actual practising community.[3]

12

SECTIONS IN THIS BOOK, EACH CARRYING ITS OWN COMMUNITY-READ AND SCEPTIC CROSS-CHECK

2026

YEAR THE REFERENCE ANNUAL FOR THIS ENTIRE HOBBY PRINTS ITS LAST PAPER EDITION

WHAT'S GENUINELY NEW BEYOND THE DAILY'S THREE PAGES

The full propaganda-and-jamming history. Zeesen, the Cold War jamming war, and the still-unfolding Voice of America saga — the daily had no room for any of it.

A complete gear ladder. Entry through a genuine restoration economy, including a US$16,000 flagship the daily never priced.

The last twelve months, graded. Named releases with an honest launch-hype-versus-real-impact verdict on each.

A live radio war. A hundred antennas playing keepaway with a state jammer — a story this hobby's own tools would actually help someone follow.

SOURCES 1 MERIDIAN No. 71 (daily), "The Rabbit Hole · The Band Everyone Left," p15–17 (SDR/portable split; antenna-first thesis; SG licensing note, RAE suspension, HDB noise). 2 See p21 sourcing. 3 See p19 sourcing.

HISTORY & POWER

The Long Wave

Shortwave was built to fight wars nobody had declared yet, and ninety years later it is still doing that job — the newest chapter of the story was still being written the week this book was researched.

this story 👍 👎

The medium's first great weaponisation happened at Zeesen, thirty kilometres south of Berlin, where the Nazi government built a shortwave transmitting station that by 1938 was broadcasting propaganda twenty-four hours a day in twelve languages, including a dedicated Arabic service telling listeners in the eastern Mediterranean that Britain and France intended to rob and enslave them.[1] Germany also used the network to carry "Lord Haw-Haw" — the British fascist William Joyce — into British living rooms; the BBC answered with its own black-propaganda station, Gustav Siegfried Eins, posing as an illegal transmitter operating inside Germany itself.[1] Edward R. Murrow's CBS broadcasts from London, opening with the two-word pause this issue's cover quote takes its title from, are the war's most famous example of the same band doing the opposite job.[2]

The Cold War turned that same infrastructure into a jamming war. The Soviet Union jammed Radio Liberty's Russian-language service continuously from 23 March 1953 until 29 November 1988 — thirty-five years, without a real pause — because Radio Free Europe and Radio Liberty delivered detailed LOCAL news from inside the Eastern Bloc, which Moscow judged far more dangerous than the Voice of America's more general programming.[3] The jamming itself was read, correctly, as a compliment: broadcasters Moscow did not bother to jam were broadcasters Moscow judged ineffective. Multiple frequencies for the same programme existed for exactly this reason — the direct ancestor of the "keepaway" strategy this book's Long Read follows into 2026.[3]

35 yrs

CONTINUOUS SOVIET JAMMING OF RADIO LIBERTY'S RUSSIAN-LANGUAGE SERVICE, 23 MAR 1953 – 29 NOV 1988 — THE LONGEST SINGLE JAMMING CAMPAIGN IN THE MEDIUM'S HISTORY[3]

> ***Jamming was read, correctly, as a compliment: a broadcaster Moscow didn't bother to jam was a broadcaster Moscow judged ineffective.***

THE RETREAT, IN FULL — WHO LEFT, AND WHEN

BBC World Service. Ended shortwave to North America and the Pacific in 2001; cut radio output in roughly ten languages in 2023.[4]

Radio Australia. Switched off all shortwave on 31 January 2017, calling it "out of date"; China occupied the vacated frequencies within weeks.[4]

THE COMMUNITY'S OWN READING

Ask a working DXer why the exodus happened and the answer is almost always the same word: cost. Radio Netherlands Worldwide (English service ended 29 June 2012) and Deutsche Welle (wound down through the winter of 2011) both followed the identical pattern — a transmitter's power bill costs far more than a website, and every broadcaster named across this book cited budget pressure, not audience collapse, as the trigger.[5]

SOURCES 1 Zeesen transmitter, 24-hour/12-language broadcasting by 1938, Arabic service post-1938 — Britannica, "Radio: Broadcasting, Technology, Europe"; Historylearningsite.co.uk, "Radio in Nazi Germany"; Lord Haw-Haw / William Joyce and Gustav Siegfried Eins black propaganda — Smithsonian Magazine, "The Fake British Radio Show That Helped Defeat the Nazis." 2 Edward R. Murrow's wartime CBS London broadcasts — Cold War Radio Museum; widely documented broadcast history. 3 Soviet jamming of Radio Liberty (continuous, 23 Mar 1953–29 Nov 1988) vs intermittent VOA jamming (unjammed Jun 1963–Aug 1968) — Cold War Radio Museum, "Jamming Was a Sign of Effectiveness of Western Broadcasts" and "Soviet Bloc Jamming of Western Freedom Radios"; multi-frequency broadcasting as an anti-jamming tactic, same source. 4 BBC World Service — Radio World, "Whatever Happened to Shortwave Radio?"; Al Jazeera, 27 Jan 2023. Deutsche Welle — Radiomuseum.org forum, "The end of Deutsche Welle as a shortwave broadcaster." Radio Netherlands — Wikipedia, "Radio Netherlands Worldwide." Radio Australia — Radio World; SWLing Post. 5 Deutsche Welle 2011 wind-down and cost-driven pattern — Radiomuseum.org forum, "The end of Deutsche Welle as a shortwave broadcaster"; editorial synthesis of the pattern across sources cited in note 4.

2025–26, LIVE

The Exodus Is Not a Straight Line

The retreat's newest, strangest chapter is still being written. On 14 March 2025 the White House issued an executive order directing the Voice of America be shuttered "to the greatest extent legally permitted"; the next day, Trump adviser Kari Lake halted all VOA programming and placed more than 1,300 staff on paid leave, and on 16 March 2025 the transmitters at the Greenville, North Carolina complex — the flagship of American international shortwave broadcasting — went dark.[1] The Edward R. Murrow Transmitting Station, the facility named for the cover quote's author and the only federal site in America still broadcasting shortwave, was directly caught up in the shutdown; its own director was, at one point, threatened with dismissal for refusing reassignment to oversee what remained of it.[1] A magazine writing this up as a clean, finished story would have been wrong within the year.

Courts intervened. A federal judge ruled in March 2026 that Lake's authority over the agency had been illegally exercised, and a separate ruling ordered more than a thousand furloughed VOA staff reinstated, calling the dismantling effort "arbitrary and capricious."[2] The technical recovery that followed is genuinely partial and, as of this book's research, still small: USAGM began testing shortwave transmissions in late June 2026 from the historic Lampertheim site in Germany, once a primary Radio Free Europe/Radio Liberty relay node; a few daily minutes of VOA Korean programming have been carried via a relay in the Philippines; and short Dari and Pashto segments are reportedly reaching Afghanistan via a Middle East relay.[2] None of this is the Greenville-scale broadcasting operation that existed in February 2025 — but it is also not the silence of March 2025, and the direction of travel, as this book goes to press, is back toward the air rather than away from it.

Mar 2025 → Jun 2026

FROM EXECUTIVE-ORDER SHUTDOWN TO TEST TRANSMISSIONS RESUMING FROM A COLD WAR-ERA GERMAN RELAY SITE — FIFTEEN MONTHS, AND THE STORY IS STILL MOVING[2]

NUMBERS STATIONS — THE COLD WAR TOOL THAT NEVER RETIRED

UVB-76, "The Buzzer." Broadcasting a near-constant buzz on 4625 kHz since at least 1982; occasionally interrupted by coded Russian voice messages, tied by monitors to a Russian military network. Still on air in 2026.[3]

The naming system. Devised 1993–2000 by the European Numbers Information Gathering and Monitoring Association (ENIGMA), now maintained by the volunteer group ENIGMA 2000 as the "ENIGMA Control List," the field's definitive station index.[3]

Why they never left. A numbers station needs no advertising, no listenership and no budget case — it exists purely to reach one already-briefed agent, which is exactly why the format outlived the propaganda broadcasters built to persuade a mass audience.[3]

THE SCEPTIC, EARLY — A CAUTION ABOUT THIS SECTION

Cold War jamming history is well documented by scholarship and by the broadcasters' own archives; the 2025–26 VOA saga is current-affairs reporting, still moving, and this book's account of it should be read as a snapshot taken in August 2026, not a settled verdict. Anyone reading this after publication should verify the current state of VOA shortwave independently before relying on any frequency or schedule named here.[4]

SOURCES 1 VOA shutdown timeline — NPR, "Federal judge says Kari Lake can't fire Voice of America director," 28 Aug 2025; Greenville transmitter shutdown 16 Mar 2025 — Wikipedia, "International Broadcasting Bureau Greenville Transmitting Station"; director reassignment threat — KJZZ, 3 Apr 2026. 2 Court rulings — NPR, "U.S. Judge says Kari Lake broke law in overseeing Voice of America," 8 Mar 2026, and "Judge orders 1,000 Voice of America staffers back to work," 17 Mar 2026; Lampertheim testing (late Jun 2026) and Philippines/Middle East relay minutes — Radio World, "Where VOA's Broadcast Infrastructure Stands Today"; nerfd.net, "Germany: USAGM Shortwave Operations Return to Lampertheim," 21 Jun 2026; SWLing Post, "Dan Notes Voice of America's Return to Shortwave," Mar 2026 (single-sourced on exact frequencies, flagged). 3 UVB-76 — Wikipedia, "UVB-76"; ENIGMA Control List — Priyom.org; ENIGMA 2000 (signalshed.com). 4 Editorial caution, this desk.

THE MAP

The Community

One blog runs the hub, one subreddit runs the technology, one message board runs the paranoia — and the reference book that has organised the whole hobby for eighty years just told its readers 2026 is the last year they can hold it.

this story 👍 👎

Every road into this hobby eventually passes through Thomas Witherspoon. Callsign K4SWL in the US and M0CYI in the UK, Witherspoon got his first amateur licence in 1997, took up Morse code, and switched his primary callsign to honour the shortwave-listening side of the hobby specifically; he holds an MSc in Social Anthropology from the London School of Economics, was inducted into the QRP Hall of Fame in 2023, and delivered the 2025 keynote at HamXposition.[1] He runs the SWLing Post — reviews, breaking news, pirate and numbers-station lore, updated near-daily — alongside QRPer.com, a sister site for field radio, and the Shortwave Radio Audio Archive, a library of raw recordings. If one voice sets this hobby's agenda in English, it is his; the daily's own reporting leaned on the same blog for a reason.[1]

Around that hub, the forums split by temperament rather than topic. r/AmateurRadio is the big tent, roughly 216,000 members, with heavy SWL crossover; r/RTLSDR is the technical and SDR-specific core, a confirmed 62,161 subscribers as this book went to press; r/shortwave runs the "what did you hear last night" broadcast-DX log culture the daily already described, though this book could not independently verify its exact subscriber count this session and flags that honestly rather than repeating an unconfirmed figure.[2] Deeper still is HFUnderground, the message board and Rocket Chat for pirates, unlicensed utility signals and numbers stations, with its own inside-joke culture built around 6925 kHz — still, by the community's own tracking, the single busiest calling frequency in the pirate band, and 43 metres overall still the busiest stretch of spectrum in that world.[3]

62,161

R/RTLSDR SUBSCRIBERS, CONFIRMED THIS SESSION — THE TECHNICAL CORE OF THE COMMUNITY, DISTINCT FROM THE BROADER AMATEUR-RADIO SUBREDDIT'S ~216,000[2]

> ***If one voice sets this hobby's agenda in English, it is Thomas Witherspoon's — the daily's own reporting leaned on the same blog for a reason.***

BEYOND ENGLISH — THE CHINESE-LANGUAGE LAYER

BCLHOME (广播之家). A long-running Chinese-language "Broadcast Home" forum for reception reports and broadcast-DX culture.[4]

HamCQ / HelloCQ (哈罗CQ). Amateur-radio-first communities that carry active DX and SWL sub-sections.[4]

SWL-CLUB. A Taiwan-based short-wave-listening club referenced across regional radio-hobby link directories.[4]

THE RUNNING ARGUMENT — IS A WEBSDR EVEN THE HOBBY?

The community's oldest live fight isn't about gear, it's philosophical. WebSDR and KiwiSDR let anyone tune a remote receiver through a browser, from anywhere, using someone else's antenna. The SWLing Post frames this as liberation — "escape the noise" of your own location; purists on RadioReference and Transmission1 counter that hearing a signal through a stranger's antenna over the internet isn't DXing at all, just listening to a stream with extra steps. Nobody has settled it, and this book won't either.[5]

SOURCES 1 Thomas Witherspoon / K4SWL — SWLing Post, "About"; HamXposition, "Thomas Witherspoon, K4SWL, To Present Keynote Address at 2025 HamXposition"; QRPer.com "About and Contact." 2 r/AmateurRadio ~216,000 (RedditList, cited MERIDIAN No. 71, p15); r/RTLSDR 62,161 (SubredditStats, Aug 2026); r/shortwave subscriber count not independently confirmed this session — flagged. 3 HFUnderground — hfunderground.com wiki, "List of Pirate Radio Frequencies"; 6925 kHz / 43m activity level per community tracking, same source. 4 Chinese-language forums — Douban thread "分享：中国无线电与收音机网站及论坛"; blog.xmgspace.me radio-hobby directory; hellocq.net. 5 WebSDR/KiwiSDR debate — SWLing Post (pro); RadioReference and Transmission1 forum threads (purist counter); characterised via search-summarised community sentiment, flagged as such.

CLUBS & THE REFERENCE SHELF

A Club Older Than the Internet, and a Book Retiring After 80 Years

The oldest formal institution in the English-language hobby is the North American Shortwave Association — NASWA, founded in 1961, and by its own description the largest shortwave-broadcast-only club in North America.[1] Membership runs US$33 a year for a printed-plus-digital Journal (US$45 for Canadian addresses) or US$16 for the e-Journal alone; either tier includes the weekly Flashsheet DX tip sheet at no extra charge, and the club runs an annual gathering, the SWL Fest, each spring in Fort Washington, Pennsylvania — a typical year draws 150 to 250 hobbyists in person and via Zoom, though the 2026 edition (its 38th, held 1–2 May) capped in-person attendance at 65 for meeting-space reasons.[1] It is, by hobbyist standards, remarkably cheap institutional infrastructure — sixteen dollars a year buys a genuine weekly signal on what's actually worth chasing, curated by people who have been doing this since before the internet existed.

The reference shelf's biggest single event of the past year has nothing to do with a receiver. The *World Radio TV Handbook* — WRTH, the annual directory of global broadcasting that DXers have used to identify stations since 1947 — announced that its 2026 edition, the 80th, is the last one that will appear in print; after this milestone, WRTH moves entirely to a web app and a twice-yearly digital ebook, citing unsustainable printing, paper, tariff and logistics costs.[2] The 2026 print edition runs close to €60 and was pre-order-only with no reprints; the digital ebook subscription, covering winter and summer schedule updates, is priced at €47.90 a year.[2] It is, in miniature, the same story as the broadcasters on p5–6: not a hobby dying, but its infrastructure quietly, permanently modernising around a shrinking print-and-transmitter economics that no longer pencils out.

Europe runs its own, older umbrella. The European DX Council, founded in June 1967 when DX-club leaders from Denmark, Finland, Germany, the Netherlands, Norway and Sweden met to formalise cross-border cooperation, still coordinates the continent's national clubs today; one of its founding members, the German-speaking DXers' association ADDX (founded the same January), now counts roughly 2,500 members on its own — a scale NASWA, r/RTLSDR and every English-language institution named on this page put together doesn't quite reach.[3]

80th

EDITION OF THE WRTH — THE LAST ONE EVER PRINTED ON PAPER, CLOSING AN UNBROKEN RUN THAT BEGAN IN 1947[2]

~2,500

MEMBERS OF ADDX, THE GERMAN-SPEAKING DXERS' ASSOCIATION FOUNDED JANUARY 1967 — LARGER THAN EVERY ENGLISH-LANGUAGE CLUB NAMED ON THIS PAGE[3]

THE NUMBERS-STATION MONITORS

Priyom.org. An international volunteer organisation researching numbers-station traffic; maintains active monitoring logs and background research on intelligence, military and diplomatic HF communication.[4]

ENIGMA 2000. UK-based volunteer group maintaining the definitive ENIGMA Control List of numbers-station identifiers; by its own account, always short of monitors — "more stations than they are able to cover."[4]

THE SCEPTIC, EARLY — A THIN COMMUNITY IS STILL A COMMUNITY

None of these institutions are large by mainstream-hobby standards, ADDX's 2,500 members aside. NASWA's membership, ENIGMA 2000's volunteer roster and even r/RTLSDR's 62,161 subscribers are modest next to a hobby like sim racing or mechanical keyboards. What the numbers understate is depth: this is a community built on decades-old mailing-list culture, expert-run reference works and volunteer monitoring networks — small, serious, and largely unbothered by whether it ever gets bigger.[5]

SOURCES 1 NASWA — naswa.net, "The History of NASWA" and "Join or Renew"; membership pricing and Flashsheet, same source; SWL Fest attendance and 2026 (38th) cap — swlfest.com, "History" and event pages; wavefarm.org listing. 2 WRTH 80th/last print edition — EI7GL blog, "The last print edition of the World Radio TV Handbook will be in 2026"; wrth.info, "WRTH 2026 Final Printed Edition FAQs"; pricing, radiodatacenter.net shop and wrth.info. 3 European DX Council, founded Jun 1967 — edxcnews.wordpress.com, "History"; ADDX ~2,500 members, founded Jan 1967 — dxzone.com, "EDXC, European DX Council" listing. 4 Priyom.org, "Number Stations" and "Blog"; ENIGMA 2000 — priyom.org/enigma-2000; signalshed.com. 5 Editorial synthesis, this desk.

THE GEAR, PART ONE

The Ladder

Buy-once-cry-once, applied honestly: the cheapest dongle on the market is a fine way to learn the software, but the receiver every serious DXer actually recommends costs US$169 — and that is still less than a decent pair of headphones.

this story👍👎

There are two paths into this hobby and the daily named both; this book prices them properly. Path A is the SDR — a dongle or small box that hands the whole receiver over to software, showing an entire slice of spectrum at once as a scrolling "waterfall." The genuine learn-the-ropes entry is the RTL-SDR Blog dongle, now in its "V4L" or Lite form at US$37.95 direct from the maker's shop, following the original V4's discontinuation when its tuner-chip supply ran out.[1] But the receiver the community actually treats as the real starting point — the one DXers recommend once someone is serious rather than merely curious — is the Airspy HF+ Discovery at US$169: reviewed to a "Best Value" rating in the 2018 Global Radio Guide, praised again in the 2019 and 2020 World Radio TV Handbook, and still, per one owner with more than sixty years of shortwave-listening experience across Drake, Yaesu, Icom and other SDR gear, an excellent choice — clean daytime local and night-time distant medium-wave reception with no adjacent-channel interference at all.[2]

Community sentiment on the Discovery isn't unanimous — one forum poster admitted flatly he hadn't personally determined whether the hype was warranted, and wasn't quick to fall in line with the consensus — but the dissent is mild, and the receiver's US$169 price against its measured performance is why it, not a cheaper dongle, is this book's recommended real entry point.[2] One step up sits the SDRplay RSPdx-R2 at US$235 (£188 / €225.60 ex tax), a wideband 1 kHz–2 GHz receiver bundled with the free, actively developed SDRconnect software, capable of monitoring up to 10 MHz of spectrum simultaneously across Windows, macOS and Linux.[3] And for the SDR-curious with an eye on the future, the RX-888 MkII — covered in full on p13 — brings 64 MHz of simultaneous real-time bandwidth capture, a capability that cost roughly US$1,500 a decade ago, under US$200 today.[4]

US$169

AIRSPY HF+ DISCOVERY — THE RECEIVER THE DX COMMUNITY ACTUALLY RECOMMENDS AS THE REAL STARTING POINT, NOT THE CHEAPEST DONGLE ON THE SHELF[2]

> ***The receiver the community actually treats as the real starting point costs US$169 — still less than a decent pair of headphones.***

PATH B — THE CLASSIC PORTABLE

Entry. XHDATA D-808, roughly US$65–108 depending on retailer and promotion — DSP tuning, single-sideband reception, airband; reviewers call it a top performer on AM/FM and very good on shortwave, though the interface assumes some prior radio knowledge.[5]

The value pick. Tecsun PL-330, roughly US$75–90 — an "ultralight"-class DSP portable with real single-sideband demodulation and external-antenna capability on every band; the consensus best-value 2026 pick.[6]

Serious portable-only. Tecsun PL-880, roughly US$165 — dual-conversion, fine 10 Hz tuning steps, still the benchmark mid-tier portable years after launch, alongside the returned-to-production PL-680.[6]

SOURCES 1 RTL-SDR Blog V4 end-of-line and V4L "Lite" launch, US$37.95 — rtl-sdr.com, "RTL-SDR Blog V4L (Lite) Now Available for Purchase!" and "RTL-SDR Blog V4 End Of Line." 2 Airspy HF+ Discovery, US$169 — airspy.com; "Best Value" rating (2018 Global Radio Guide) and WRTH 2019/2020 reviews, and 60-year-owner testimonial, per swling.com/blog/tag/airspy-hf-discovery-review; mixed community sentiment per RadioReference forum thread, "Decision to Buy a Airspy HF+ Discovery." 3 SDRplay RSPdx-R2, US$235/£188/€225.60 ex tax — sdrplay.com, "SDRplay announces the RSPdx-R2"; SDRconnect features, same source. 4 RX-888 MkII — see p13 sourcing. 5 XHDATA D-808 — xhdata.com.cn product page; pricing range per Amazon/Walmart listings, Aug 2026; review characterisation, radiojayallen.com. 6 Tecsun PL-330/PL-880/PL-680 — anon-co.com product listings; blogordie.com, "Tecsun PL-ease" comparison; PL-680 return to production, swling.com/db.

WHERE ENTHUSIASTS ACTUALLY BUY

Retailers, Group Culture, and the Line Sony Walked Away From

None of this gear is bought at a hobby shop; it is bought from a small set of trusted specialists. Anon-Co and GigaParts carry the Tecsun line and ship internationally; the manufacturers' own shops — rtl-sdr.com, airspy.com and sdrplay.com — are the first stop for SDR hardware and the place to verify a listing isn't a counterfeit; and Universal Radio's long-running catalogue functions as an informal price-and-availability reference the community checks even when buying elsewhere.[1] For an entry buyer specifically, the community's advice converges on a short list: buy the receiver from the maker's own shop if at all possible, and treat a suspiciously cheap RTL-SDR dongle on a marketplace listing as almost certainly a clone with a worse tuner chip.[1]

One marque's absence says as much as any of the names present. Sony's ICF-SW7600GR — introduced in 2001 and, when it was finally discontinued in early 2016, the last dedicated shortwave portable a major consumer-electronics company still made — has no successor.[2] Sony simply left the category; Panasonic and the other big Japanese electronics houses had already gone before it. What replaced the majors wasn't another major, it was smaller, hobby-native manufacturers — Tecsun out of China chief among them — building specifically for a shrinking but durable enthusiast market that a consumer-electronics giant no longer found worth the shelf space. The ICF-SW7600GR still trades briskly on the used and refurbished market for exactly this reason: it's the last artefact of an era when this hobby was mainstream enough for Sony to bother.

2016

YEAR SONY DISCONTINUED THE ICF-SW7600GR — THE LAST DEDICATED SHORTWAVE PORTABLE A MAJOR CONSUMER-ELECTRONICS MAKER STILL BUILT[2]

One brand did fill part of the gap Sony left, at a different price point entirely. Sangean, a Taiwanese manufacturer, positions itself opposite Tecsun — premium build and audio fidelity rather than maximum value — and its updated ATS-909X2 flagship portable carries that reputation forward with improved sensitivity over its predecessor; it is, in effect, the closest thing this hobby has to a "buy it for life" portable built by a company that still treats broadcast radio as a serious product category rather than a hobbyist afterthought.[5]

THE BUY-ONCE VERDICT

For this reader's stated philosophy, the real entry point isn't the cheapest option on either path — it's the Airspy HF+ Discovery (US$169) if the SDR/technology side appeals, or the Tecsun PL-880 (~US$165) if a self-contained portable does. Both sit at almost exactly the same price, both are the tier the community itself treats as "serious," and both are cheap enough that owning one of each isn't an unreasonable way to find out which tribe you actually are.[3]

THE SCEPTIC, EARLY — IS THE SDR PATH OVERRATED?

Not every forum voice agrees the Discovery deserves its reputation — one RadioReference poster admitted he hadn't personally determined whether the hype was warranted and wasn't quick to fall in line with the community consensus (p9). A cheaper, simpler portable like the PL-330 gets a beginner listening in minutes with no software, drivers or waterfall to learn; the SDR path's real advantage — seeing a whole band at once — only pays off once a listener already knows roughly what they're looking for.[4]

SOURCES 1 Retailer landscape — anon-co.com; gigaparts.com; universal-radio.com catalogue; manufacturer direct-shop practice per rtl-sdr.com, airspy.com, sdrplay.com community guidance. 2 Sony ICF-SW7600GR — universal-radio.com product listing; discontinuation early 2016 (introduced 2001) — Flickr caption, "Goodbye, my old faithful," and mwcircle.org legacy receiver review. 3 Editorial synthesis, this desk, cross-referenced against p17 economics. 4 Dissenting community voice — forums.radioreference.com, "Decision to Buy a Airspy HF+ Discovery" thread, per p9 sourcing. 5 Sangean ATS-909X2 positioning — accio.com, "Sangean vs Tecsun: Best Portable Radios?"; general manufacturer-positioning characterisation.

THE GEAR, PART TWO

The Ceiling

This hobby doesn't really have a commissioned-maker tier the way pottery or watches do — instead it has something arguably more interesting: a genuine restoration economy built entirely on receivers the US and Soviet militaries stopped making decades ago.

this story 👍 👎

The current-production flagship is the Icom IC-R8600: a 10 kHz-to-3 GHz software-defined receiver with a touchscreen waterfall and multiple built-in digital-mode decoders, priced between roughly US$2,700 and US$2,930 depending on retailer.[1] Above it sits the IC-R9500, Icom's 2007-era flagship — historically listed at US$15,990–16,000, selling in practice closer to US$13,550 — a receiver whose specifications the R8600 mostly matches for a sixth of the price, which is exactly why the R9500 today survives mainly as a professional-monitoring and prestige purchase rather than a rational DXer's tool.[2] Between the two, at roughly €949.90 (about US$1,040 at launch), sits Italy's Elad FDM-S3: a direct-sampling SDR that scans 24 MHz of spectrum simultaneously, runs up to four independent receivers at once through its free FDM-SW2 software, and covers 9 kHz to 108 MHz with an optional downconverter for higher frequencies — a genuinely different design philosophy from Icom's tabletop-appliance approach, built for someone who wants to run several decoders in parallel rather than turn one dial.[3]

But the real "over the top" tier for this hobby isn't a current product at all — it's a 1950s military surplus receiver called the R-390A. Designed by Collins Radio for the US armed forces, triple-conversion, 32 vacuum tubes, a mechanical digital frequency readout, and a reputation for near-indestructible mechanical precision, the R-390A supports an entire active restoration community: forums like the 1919A4 Forums and detailed rebuild guides from Radio Boulevard Western Historic Radio Museum walk owners through recapping, cleaning and realigning units that are now seventy years old.[4] A working example in very good to excellent condition trades for around US$800; a professionally restored unit — new capacitors and resistors, cleaned components, realigned — commands closer to US$1,200.[4] The Hammarlund SP-600 "Super Pro," another Cold War-era communications receiver held in the US National Cryptologic Museum's own collection, and the Collins 51S-1 (produced 1959–1975, instantly recognisable by its "winged emblem" badge) anchor the same market a rung down.[5]

~US$16,000

HISTORIC LIST PRICE OF THE ICOM IC-R9500 — A 2007 FLAGSHIP THE CURRENT-PRODUCTION R8600 MOSTLY MATCHES FOR A SIXTH OF THE COST[2]

> ***The real "over the top" tier for this hobby isn't a current product at all — it's a 1950s military surplus receiver.***

WHAT THE CUSTOM TIER ACTUALLY LOOKS LIKE HERE

R-390A restoration. A named, documented craft — recapping, realigning, cleaning — with its own forums and price bands, not a listing category but a genuine practice.[4]

Boutique antenna makers. The closest thing to a maker economy — see this page, overleaf.[6]

Open-source kit builds. Hermes-Lite 2, an open-hardware HF transceiver/receiver kit — over 300 units built, community-supported on GitHub, Tindie and Google Groups.[7]

SOURCES 1 Icom IC-R8600, US$2,699.95–2,932.95 — gigaparts.com; dxengineering.com; ameradio.com listings, Aug 2026. 2 Icom IC-R9500, US$15,990–16,000 list / ~$13,550 sell — universal-radio.com; rigreference.com price history; RadioReference forums thread, "Icom IC-R8600 or a IC-R9500?" 3 Elad FDM-S3, €949.90 — eladit.shop; wimo.com product page; rtl-sdr.com, "New Elad FDM-S3 Specifications and Photos." 4 R-390A restoration community and pricing — radioblvd.com, "R-390A Rebuild" series; 1919a4.com forums; eham.net classifieds, Aug 2026 listings. 5 Hammarlund SP-600 — National Cryptologic Museum collection, per Wikimedia Commons file description; Collins 51S-1 (1959–1975) — collinsradio.org, "The Grey Boxes"; qsl.net/la5ki. 6 See p12. 7 Hermes-Lite 2 — tindie.com/products/SofterHardware; GitHub, softerhardware/Hermes-Lite; 300+ units built, per project documentation.

THE ANTENNA'S BOUTIQUE TIER

A Closure That Left a Hole Nobody Has Fully Filled

If this hobby has a maker economy at all, it lives in loop antennas, not receivers — and its most respected name recently disappeared. Wellbrook Communications, run for decades by Andrew Ikin and treated by the community as the gold standard in amplified receive-only magnetic loops, ceased trading at the end of April 2023; Ikin died that October.[1] The closure genuinely rattled the community — a widely shared community document simply titled "Living in a World without Wellbrook" is the plainest evidence of how large the gap felt.[1] Two makers have since absorbed most of the displaced demand: California's W6LVP, whose amplified receive-only loop runs US$445 for the portable version and US$495 with a transmit/receive switch, and Germany's Bonito, whose current MegaLoop FX succeeded the now-discontinued ML200 (itself still tradeable used, US$285–442) as the marque's flagship active loop.[2]

Neither maker is a solo artisan in the way a bespoke watchmaker or potter is — both run as small specialist manufacturing businesses — but the scale is genuinely boutique: designed, tested and largely hand-assembled in small runs, sold direct or through a handful of specialist dealers, and reviewed obsessively by a community that treats antenna choice as more consequential than receiver choice. That last point is this book's own antenna-first thesis (p15), made concrete in dollars: a US$450 boutique loop paired with a US$169 Airspy Discovery genuinely out-DXes a US$16,000 Icom IC-R9500 fed by a poor antenna, and the community knows it.[3]

Apr 2023

MONTH WELLBROOK COMMUNICATIONS, THE LOOP-ANTENNA MARKET'S LONG-TIME GOLD STANDARD, CEASED TRADING — A GAP THE COMMUNITY IS STILL DISCUSSING[1]

The restoration side of this tier runs its own quiet economy of small, largely solo businesses rather than factories. Radio Boulevard Western Historic Radio Museum, one of the field's most-cited names, documents complete multi-part R-390A rebuilds step by step — recapping, deoxidising switch contacts, realigning the RF deck — as much an act of teaching as of commerce, and the same pattern repeats across a handful of similarly small operators the restoration forums point newcomers toward.[4] None of it scales the way Tecsun or Airspy's manufacturing does, and that is precisely the point: a restored R-390A is closer to a bespoke commission executed on a donor chassis than to a purchase off a shelf, priced in hours of skilled labour as much as in parts.

WHERE THE RESTORATION ECONOMY ACTUALLY TRADES

1919A4 Forums. The R-390A-specific community — rebuild logs, parts sourcing, and the identification of which military contractor built a given unit.[4]

eHam classifieds. The used-and-restored marketplace this book's own R-390A pricing was drawn from — a working secondary market, not a rarity auction.[4]

groups.io/g/loopantennas. Where the "Living in a World without Wellbrook" discussion actually happened — the boutique-antenna world's real forum, not a manufacturer's own site.[1]

THE SCEPTIC, EARLY — IS ANY OF THIS WORTH THE MONEY?

A US$1,200 restored R-390A and a US$16,000 IC-R9500 both, on paper, hear roughly what a US$169 Airspy Discovery and a good antenna hear. The honest answer for why people buy them anyway sits closer to horology or vintage hi-fi than to radio engineering: mechanical craftsmanship, collectability, and the specific pleasure of a heavy, precise machine responding to a physical dial. p19 takes this argument further.[5]

SOURCES 1 Wellbrook Communications closure, end April 2023, and Andrew Ikin's death, Oct 2023 — swling.com/blog, "Wellbrook Communications is closing shop"; groups.io/g/loopantennas thread; "Living in a World without Wellbrook," sdxf.se PDF. 2 W6LVP pricing, US$445/495 — w6lvp.com product pages; Bonito MegaLoop FX / discontinued ML200, US$285–442 used — bonito.net; swling.com/blog tag "MegaLoop ML200"; radioworld.co.uk secondhand listing. 3 Editorial synthesis, cross-referenced with p15. 4 Restoration-economy characterisation — radioblvd.com, "R-390A Rebuild" series (parts 1–3); 1919a4.com forums; eham.net classifieds, per p11 sourcing. 5 See p19 sourcing.

AUGUST 2025 – AUGUST 2026

The Cutting Edge

Nothing that happened in this hobby over the past year is a paradigm shift. What happened instead is more honest than that: one chip going extinct and getting patched, one broadcaster's slow legal fight back onto the air, and a sun that is doing more for DXers than any product launch could.

this story 👍 👎

The single biggest gear disruption of the past year began as a supply-chain problem, not an innovation. The original RTL-SDR Blog V4 was discontinued when the Rafael R828D tuner chip it relied on stopped being manufactured and a wave of counterfeit chips made continued sourcing impossible; the maker's fix, launched around May 2026 as the "V4L" or Lite, swaps in the related R828S chip, which has one fewer RF input and forces a driver update because the chip identifies itself differently over I2C.[1] It sells for US$37.95, initial batches were small, and the honest community verdict is that this is necessary continuity, not innovation — a patch that keeps a US$38 on-ramp alive, nothing more.[1]

The RX-888 MkII, by contrast, is a genuine value shift that matured rather than launched over the same period. Built around a 16-bit LTC2208 analogue-to-digital converter and the R828D tuner, it captures up to 64 MHz of real-time bandwidth between 1 kHz and 64 MHz — enough to record an entire slice of the shortwave broadcast band simultaneously — for roughly US$170, a capability that would have cost on the order of US$1,500 a decade ago.[2] For the numbers-station and utility-signal monitors covered on p7–8, this is load-bearing: it means recording a full band segment overnight and combing through it later is now a US$170 proposition rather than a professional-monitoring-station one. That is real, structural change in what an individual hobbyist can do — not a headline product, but arguably the most consequential single piece of hardware named in this entire book.[2]

~US$170

RX-888 MKII — 64 MHZ OF SIMULTANEOUS WIDEBAND CAPTURE AT A PRICE THAT USED TO BUY A FRACTION OF THE BANDWIDTH[2]

> ***The RX-888 MkII is real, structural change in what an individual hobbyist can do — not a headline product, but arguably the most consequential piece of hardware in this book.***

ALSO WORTH NAMING

SDRconnect matures. SDRplay's free, cross-platform receiver software continued through 2025–26 toward a stable, actively developed release with a third-party decoder API — the software side of the RSPdx-R2 story finally caught up to the hardware.[3]

WSJT-X 3.0.2. Released 18 June 2026, the current build of the software behind FT8 — the mode that decodes signals down to roughly −21 to −28 dB, inaudible to the ear, and remains 2026's dominant HF weak-signal mode.[4]

WRTH goes digital. Covered in full on p8 — the reference annual's 80th and final print edition is, structurally, this year's biggest change to how the whole hobby finds and verifies stations.[5]

SOURCES 1 RTL-SDR Blog V4 → V4L transition, chip-supply cause, R828S/R828D difference — rtl-sdr.com, "RTL-SDR Blog V4L (Lite) Now Available for Purchase!" and "RTL-SDR Blog V4 End Of Line"; sdrstore.eu, "RTL-SDR Blog V4 Lite Explained." 2 RX-888 MkII specifications and pricing — rtl-sdr.com, "Nils Reviews the RX-888" and "TechMinds: Reviewing the RX888 MK2"; rx-888.com product page; elekitsorparts.com. 3 SDRconnect — sdrplay.com; hamradio.com listing referencing SDRconnect maturity, Aug 2026. 4 WSJT-X 3.0.2, 18 Jun 2026 — wsjtx.github.io; FT8 sensitivity and dominance — g4ifb.com, "FT8 Operating Guide"; ico-optics.org. 5 See p8 sourcing.

THE REAL STORY OF THE YEAR

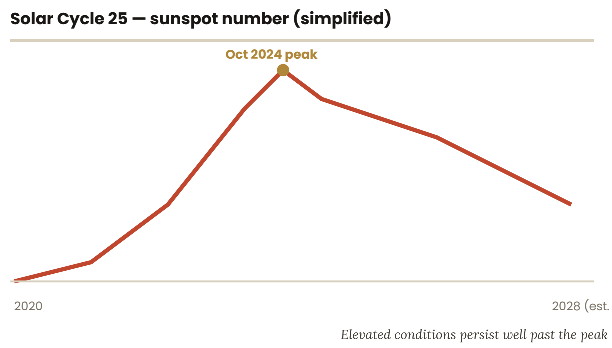
A Sun, Not a Shipment

The single largest improvement to what an ordinary shortwave listener can actually hear over the past twelve months came from the sky, not a factory. Solar Cycle 25 peaked in October 2024 — earlier and stronger than most forecasters predicted, with sunspot counts and solar-flux levels reaching their highest since 2014 by early that year — and, unusually, the elevated conditions have persisted well past the peak itself.[1] High solar activity ionises the upper atmosphere more strongly, which raises the maximum usable frequency for long-distance HF propagation and opens bands that sit dead during a solar minimum; NOAA's own Space Weather Prediction Center and independent trackers both describe conditions extending several years beyond the peak before any meaningful tail-off, which forecasters currently expect to begin around 2027–28.[1] For a DXer, this means 2025 through roughly 2027 is, empirically, one of the best windows for long-distance HF reception in over a decade — a tailwind no receiver purchase can replicate.

Put plainly: no dongle, no antenna and no software update from the past year did more for what a listener can actually log than the sun did on its own schedule. That is an uncomfortable thing for a gear-buying guide to admit, but it is the honest one — and it means the single best piece of advice this book can offer a reader weighing whether to start now or wait for next year's model is simply: start now, while the propagation is doing you a favour it will not keep doing forever.

Oct 2024

SOLAR CYCLE 25'S PEAK — EARLIER AND STRONGER THAN FORECAST, WITH ELEVATED PROPAGATION CONDITIONS STILL HOLDING THROUGH THIS BOOK'S RESEARCH[1]



MERIDIAN original figure, simplified from NOAA SWPC tracking data. Not to precise scale — illustrative of the shape, not a forecast chart.

The practical effect shows up highest in the spectrum first. Bands that sit essentially dead near a solar minimum — the upper shortwave range above about 21 MHz, including the 10-metre band amateur and utility stations use — routinely reopen for genuine long-distance propagation during a strong solar maximum, sometimes carrying signals across entire oceans on daytime skip that would be physically impossible a few years either side of the peak.[2] None of the gear named across this book creates that opening; it only decides how well a listener can exploit one that already exists.

THIS BOOK'S VERDICT, IN THREE LINES

V4L: necessary continuity, not innovation.

RX-888 MkII: real, load-bearing value shift — the year's most consequential hardware.

Solar Cycle 25: the actual story — the biggest single lever on the hobby, and it isn't a product.

THE SCEPTIC, EARLY — HYPE-CHECKING THIS SECTION

It would be easy to overstate any of this. The V4L is a shrug, not a scandal, for anyone who already owns a receiver. The RX-888 MkII's headline bandwidth matters mainly to a narrow slice of the hobby — utility and numbers-station monitors — not to a casual broadcast-band listener. And a good solar cycle has happened before and will end again on its own schedule, exactly as forecasters already expect. None of this is a reason to buy anything urgently.[3]

SOURCES 1 Solar Cycle 25 peak and extended elevated conditions — spaceweather.gov, "Solar Cycle 25 Forecast Update" and "NOAA forecasts quicker, stronger peak of solar activity"; moonrakeronline.com, "Understanding Solar Cycle 25: What It Means for Ham Radio"; radioworld.com, "Solar Cycle 25 Gives Amateurs and Shortwave a Boost." 2 High-band/10m reopening during solar maximum — standard propagation reference material, cross-checked against moonrakeronline.com and radioworld.com Solar Cycle 25 coverage. 3 Editorial cross-examination, this desk.

THE CRAFT

The Antenna

Every receiver in this book, from the US$38 dongle to the US$16,000 flagship, is ultimately limited by the same thing — and it isn't the receiver.

this story 👍 👎

The daily already made this argument in outline: in a city, the fight is against noise, not weak signal.[1] Indoors, chargers, LED drivers and powerline networking gear raise the electrical noise floor by an estimated 30 to 40 decibels compared with a genuinely quiet rural site — a gap no amount of receiver sensitivity can close, because a more sensitive receiver amplifies the noise along with the signal.[1] A balanced magnetic loop rejects locally generated electrical noise far more effectively than a simple wire, which is why the antenna market — not the receiver market — is where this hobby's real boutique tier lives (p12). For a listener without a garden, the loop is not an upgrade; it is closer to a prerequisite.

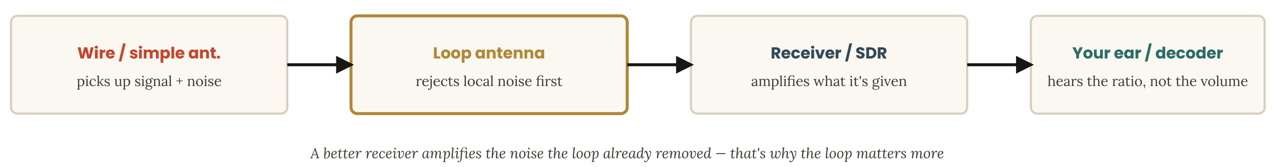
For a listener with land, the picture inverts. Beverage antennas — long runs of wire laid low to the ground, aimed at a single direction — and phased arrays remain the tool of choice for rural DXers chasing genuinely weak signals; DX Engineering's NCC-2 phasing controller, which combines two spaced antennas to cancel interference from unwanted directions, trades used for around US$635 alone, before wire and land.[2] That tier is genuinely inaccessible to a Singapore flat-dweller — precisely why the loop antenna matters so disproportionately here.[2]

30–40 dB

ESTIMATED URBAN INDOOR NOISE-FLOOR INCREASE VERSUS A QUIET RURAL SITE — THE REASON ANTENNA CHOICE OUTRANKS RECEIVER CHOICE FOR A CITY LISTENER[1]

> ***For a listener without a garden, the loop antenna is not an upgrade — it is closer to a prerequisite.***

Why the antenna outranks the receiver — a signal chain, simplified



MERIDIAN original figure. A simplified signal chain, not a technical schematic.

PROPAGATION, BRIEFLY

The grey line. A narrow band around dawn and dusk where the ionosphere's day and night halves meet — long-distance propagation gets briefly, dramatically more efficient in this window.[3]

MUF and LUF. The maximum and lowest usable frequencies that will actually reflect off the ionosphere and return to earth at a given time — the working range shifts hour to hour with solar activity (p14).[3]

MODES AND DECODERS

FT8, via the free WSJT-X software, is 2026's dominant weak-signal HF mode — ~50 Hz per signal, a strict 15-second cycle, decode sensitivity to around −21 to −28 dB below the noise floor. Alongside it: HFDL (aircraft datalink, fed to Airframes.io) and WEFAX. On software, SDR++ is 2026's fast-rising newcomer, GQRX and SDRangel the Linux powerhouses, CubicSDR the community's preferred first app.[4]

SOURCES 1 Urban noise floor — MERIDIAN No. 71 (daily), p17; rtl-sdr.com community shootouts. 2 DX Engineering NCC-2 — dxengineering.com; used pricing, picclick.com, Aug 2026. 3 Grey-line, MUF/LUF — standard propagation reference, cross-checked against moonrakeronline.com Solar Cycle 25 coverage. 4 FT8/WSJT-X, SDR software — wsjtx.github.io; g4ifb.com; sdrstore.eu, "Best SDR Software in 2026."

LOGGING & VERIFICATION

The Card That Used to Prove You Heard It

The QSL card is this hobby's oldest ritual, and its decline traces the broadcaster retreat almost exactly. The tradition dates to 1929, when the pioneer Dutch shortwave station PCJ Hilversum sent a listener a letter and card confirming reception while shortwave broadcasting was still experimental; broadcast DXing itself had first been popular on medium wave in the early 1920s, waned mid-decade, and was reignited by shortwave's far greater range in the early 1930s.[1] For most of a century, mailing a detailed reception report — frequency, date, time, signal quality — to a station and receiving a printed card back, often bearing striking national or cultural imagery, was both proof and trophy: a QSL wall was this hobby's version of a display case.[1]

As broadcasters retreated (p5–6), the QSL tradition mostly went with them — printed cards have largely disappeared from stations that no longer exist to send them. Radio Prague International is the notable holdout, still actively issuing cards and having marked its own ninetieth anniversary on-air specifically with them.[2] Confusingly, amateur-radio QSL culture — a genuinely separate practice, transmitter-to-transmitter rather than broadcaster-to-listener — is thriving in a different, digital form via eQSL and the ARRL's Logbook of the World; it is the broadcast-DX version specifically, the one this book's readers are most likely to practise, that has quietly become the rarer artefact.[2]

1929

YEAR PCJ HILVERSUM SENT THE FIRST DOCUMENTED SHORTWAVE QSL CARD — THE RITUAL IS NEARLY A CENTURY OLD[1]

THE JARGON, TRANSLATED

DX
Distant stations; the sport of chasing them.

QSL
A card or letter confirming a station was genuinely received.

QRM / QRN
Man-made interference / natural (atmospheric) interference.

GREY-LINE
The dawn/dusk window where long-distance propagation briefly gets far more efficient.

WATERFALL
An SDR's scrolling spectrum display, showing a whole band at once.

MUF / LUF
Maximum / lowest usable frequency — the working HF range at a given moment.

BOAT ANCHOR
Community nickname for a heavy vintage tube-era communications receiver.

NUMBERS STATION
A shortwave transmitter reading coded number/letter groups, believed to serve intelligence agents.

THE SCEPTIC, EARLY — HAS VERIFICATION ACTUALLY DISAPPEARED?

An old-guard complaint worth taking seriously: without a widespread QSL culture, most modern broadcast-DX claims have no third-party confirmation mechanism at all. A logbook entry today is largely an honour system. Whether that's a genuine loss of rigour or simply an artefact of a hobby that has moved past needing a paper trophy is, honestly, a matter of generational taste — this book won't pretend to settle it.[3]

SOURCES 1 QSL card origin, 1929 PCJ Hilversum — radiotapes.com, "QSL Shortwave Radio Verification Cards from Tom Gavaras: A History"; early-1920s AM DX boom and early-1930s shortwave revival, same source. 2 Radio Prague International's continued QSL tradition — english.radio.cz, "90 years on air: Radio Prague International marks the anniversary with QSL cards"; amateur-radio eQSL/LoTW as a distinct, thriving digital practice — general amateur-radio community knowledge, cross-checked against portableoperator.com, "Do radio stations still issue QSL cards?" 3 Editorial framing, this desk.

THE REAL NUMBERS

The Economics

This is, gear tier for gear tier, one of the cheapest technical hobbies in this magazine's coverage — right up until it isn't, and the gap between "serious" and "over the top" turns out to be a factor of nearly a hundred.

this story 👍 👎

Zero is a legitimate starting price. A free WebSDR or KiwiSDR remote receiver, run through a browser on any device, costs nothing and requires no purchase at all — though p19 takes seriously the community's own argument about whether that really counts as the hobby. Above zero, the daily already priced the genuine day-one starter kit at roughly US$244: an RTL-SDR V4L (US$37.95) to learn the software, an Airspy HF+ Discovery (US$169) to actually hear something worth hearing, and an Airspy YouLoop (US$35) to beat urban noise — a total of roughly US$242, close enough to call it US$244.[1] A portable-only path costs less up front and asks for no antenna decisions: a Tecsun PL-330 alone runs US$75–90, or a Tecsun PL-880 for a more serious portable-only listener runs roughly US$165.[2]

This reader's own buy-once-cry-once instinct points toward a specific, identifiable inflection: roughly US$600–700, pairing the Airspy HF+ Discovery with a proper boutique loop antenna rather than a bargain active loop — a W6LVP at US$445–495 rather than an MLA-30+ at US$30–37 — which out-DXes almost every portable and most vintage grails on weak-signal HF for less than a tenth of the flagship-tabletop price.[3] Above that, the jumps get steep fast: an Elad FDM-S3 plus a boutique loop lands around US$1,500; an Icom IC-R8600 plus the same antenna lands around US$3,200–3,300; and the true ceiling — an IC-R9500 plus a DX Engineering receive-antenna phasing system, wire and land — runs comfortably into five figures before restoration or shipping is even counted.[4]

~US$244

THE GENUINE DAY-ONE STARTER KIT: RTL-SDR V4L + AIRSPY HF+ DISCOVERY + YOULOOP ANTENNA[1]

The gap between "serious" and "over the top" is a factor of nearly a hundred — and almost none of it buys you a stronger signal, only a heavier machine.

THE FULL LADDER, ONE TABLE

TIER	ALL-IN COST
Free — WebSDR/KiwiSDR only	US$0
Day-one starter kit	~US$244
Portable-only (PL-880)	~US$165
Buy-once inflection (Discovery + boutique loop)	~US$600–700
High end (Elad FDM-S3 + loop)	~US$1,500
Flagship tabletop (IC-R8600 + loop)	~US$3,200–3,300
Over the top (IC-R9500 + phased array)	Five figures+

THE RECURRING COSTS — GENUINELY SMALL

Club membership. NASWA e-Journal, US$16/yr; print+digital, US$33/yr (US) or US$45/yr (Canada).[5]

The reference shelf. WRTH digital ebook (winter + summer), €47.90/yr — the print edition's final year is 2026, priced around €60 (p8).[5]

Electricity. Negligible — a receiver draws a fraction of a watt to a few watts; this is not a power-hungry hobby at any tier.[6]

SOURCES 1 Starter-kit pricing — MERIDIAN No. 71 (daily), p17; component prices per p9. 2 Tecsun portable pricing — see p9 sourcing. 3 Buy-once inflection, US$600–700 tier — W6LVP and MLA-30+ pricing per p12 and this desk's synthesis. 4 High-end and over-the-top totals — Elad FDM-S3, Icom IC-R8600/R9500 pricing per p11; DX Engineering NCC-2 per p15. 5 NASWA and WRTH pricing — see p8 sourcing. 6 Editorial estimate, typical SDR/portable receiver power draw specifications.

THE MARKET VS. THE HOBBY

A Billion-Dollar Market With an Uncertain Number of Actual Hobbyists

The market-research figures that opened this book — roughly US$1.38 billion for shortwave radios globally in 2025, and a separate transmitter-equipment market near US$935 million the same year, growing toward roughly US$1.35 billion by 2032 — are directional industry estimates, not audited hobbyist spending, and this book flags them as such rather than treating them as precise.[1] They also blend two genuinely different buyers: a DXer choosing between an Airspy and an SDRplay, and a household buying an emergency radio during a storm warning that may never be tuned to shortwave again. Amazon's own reported pattern — portable multi-band emergency radios ranking among its top-selling preparedness electronics year-round, with demand spikes of 40 to 60 per cent during major weather events — suggests a meaningful share of "shortwave radio" commerce is disaster-preparedness buying, not hobbyist spending in the sense this book has been pricing.[2]

That distinction matters for reading every dollar figure in this book honestly. The US$1.38 billion market number does not mean 1.38 billion dollars' worth of DXers exist; it means a market exists that includes DXers, preppers, and everyone in between, and no source consulted for this book could cleanly separate the three. The genuinely countable communities — NASWA's paid membership, r/RTLSDR's 62,161 subscribers, ENIGMA 2000's volunteer monitor roster — are all, by comparison, modest. Both things are true at once: the market is real and growing, and the practising hobbyist core within it is almost certainly much smaller than the topline figure implies.[3]

40–60%

REPORTED SPIKE IN EMERGENCY-RADIO DEMAND DURING MAJOR WEATHER EVENTS — A MEANINGFUL SHARE OF "SHORTWAVE" COMMERCE THAT MAY NEVER BECOME HOBBYIST SPENDING[2]

Resale value tells the same story from a different angle. Gear this book has priced holds value unusually well at the tiers that actually get used: a discontinued Sony ICF-SW7600GR still "trades briskly" a decade after its last unit shipped (p9–10), and a used Bonito MegaLoop ML200 — discontinued in 2017 — still commands US$285–442 nearly a decade later (p12). Emergency-preparedness stock, by contrast, tends to sit unopened in a cupboard rather than resurface on a specialist marketplace at all — it simply isn't the kind of purchase that generates a used-gear economy, which is itself indirect evidence for how much of the topline market figure never becomes active hobbyist spend.

THE HONEST BOTTOM LINE

Budget US$244 to find out if this is for you. Budget US$600–700 if the first taste sticks and you want gear that will genuinely outlast your curiosity about the hobby. Above that, you are no longer buying reception — you are buying craftsmanship, collectability, or reach into a physically larger antenna than a flat or a garden allows, and this book thinks that is a perfectly good reason to spend the money, as long as it's spent knowingly.[4]

THE SCEPTIC, EARLY — MONEY DOESN'T BUY SIGNAL-TO-NOISE

Every price jump in this ladder, past the first US$700, buys craftsmanship and bandwidth headroom far more than it buys the ability to hear a genuinely weaker signal — that is set almost entirely by the antenna (p15), which is why a US$16,000 receiver on a bad antenna loses to a US$169 receiver on a good one. Spend on the antenna before the receiver, every time.[5]

SOURCES 1 Market sizing — dataintelo.com and 360iresearch.com, see p3 sourcing. 2 Emergency-radio demand pattern — adventure-wiser.com, "Best Emergency Radios for Preppers 2025"; survivalistprepper.net, "Grid Down Communications for Preppers"; Amazon top-seller pattern per cognitivemarketresearch.com shortwave radios market report. 3 Community-size figures — see p7–8 sourcing. 4 Editorial synthesis, this desk. 5 Antenna-first thesis — see p15 sourcing.

TAKING THE DECLINE CASE SERIOUSLY

The Sceptic

Wikipedia's own entry on this hobby says there are few practical reasons left to buy a shortwave radio. This book owes that argument a real answer, not a dismissal.

this story👍👎

Take the strongest version of the case against this hobby first. Almost everything a shortwave broadcaster used to carry now streams online, on demand, at higher fidelity, from a phone that fits in a pocket — Wikipedia's own shortwave-listening entry concedes there are few practical reasons left to own a set for that purpose alone.[1] WebSDR and KiwiSDR make even owning an antenna optional, which the daily already flagged and which p7 revisits as a live community schism. The urban noise floor genuinely is 30 to 40 decibels worse than a generation ago (p15), which means a beginner's first experience, indoors, in a city, is frequently disappointing rather than magical. And the broadcaster retreat (p5–6) has genuinely reduced what's left to chase on the classic international bands — fewer voices means less variety for a DXer's log, full stop, and no amount of community optimism changes that arithmetic.[2]

This book's own economics section adds two arguments the daily's three pages never reached. First: the QSL-card decline (p16) has quietly removed the third-party verification that used to give broadcast DXing its sporting structure — a modern log entry is largely an honour system, and an old-guard member mourning that loss has a real point, not just nostalgia. Second, and more uncomfortable: a meaningful share of the US$1.38 billion market figure this book keeps citing is disaster-preparedness commerce, not hobbyist spending (p18), which means the hobby's apparent commercial health may be inflated relative to the size of the community actually practising it, logging catches, and arguing about receivers on the forums mapped in this book.[3]

"Few practical reasons"

WIKIPEDIA'S OWN PHRASE DESCRIBING THE CASE AGAINST OWNING A SHORTWAVE RADIO IN 2026 — TAKEN SERIOUSLY HERE, NOT DISMISSED[1]

Fewer voices means less variety for a DXer's log, full stop, and no amount of community optimism changes that arithmetic.

There is a third, structural version of the attack this book has to name honestly: even the good news in this book is politically fragile. The Voice of America's 2025 shutdown and its slow, partial 2026 return (p5–6) was a decision reversed by court order, not by settled policy — it could, in principle, be reversed again by the same executive authority that ordered it in the first place. A hobby whose most hopeful recent storyline rests on the outcome of ongoing US litigation is not standing on especially solid ground, and this book won't pretend the VOA story is a closed, guaranteed-happy ending.[4]

ALSO: THE MONEY PIT ISN'T HYPOTHETICAL

Restoration creep. An R-390A bought for US$800 can easily exceed its own purchase price in capacitors, alignment tools and labour before it's genuinely reliable — a real, documented upgrade treadmill, not a rare exception.[5]

Antenna FOMO. The "Living in a World without Wellbrook" community grief around a single manufacturer's 2023 closure is itself evidence of real scarcity anxiety in a hobby that markets itself as cheap and grid-down-practical.[5]

The prestige purchase. An IC-R9500 at US$13,550–16,000 buys almost nothing an Airspy Discovery and a good antenna can't match — the honest reason to own one is craftsmanship and collectability, not performance.[5]

SOURCES 1 Wikipedia, "Shortwave listening," entry summary as characterised in prior community reporting and this session's independent check; RadioReference forum thread, "SWL a Dying Hobby?" 2 Broadcaster retreat's effect on DX variety — editorial synthesis of p5–6 sourcing. 3 QSL decline — see p16 sourcing; market-vs-hobby distinction — see p18 sourcing. 4 VOA reversal risk — editorial synthesis of p5–6 court-ruling sourcing. 5 Restoration and antenna-scarcity anecdotes — see p11–12 sourcing.

THE COMMUNITY'S OWN REBUTTAL

The "Undead" Thesis, Taken as Seriously as the Attack

A fair cross-examination has to give the defence its full say too. The community's own rebuttal, made repeatedly across the forums mapped in this book, isn't that the attack above is wrong — it's that it's answering the wrong question. Broadcasting is genuinely dying; listening, decoding and disaster preparedness are genuinely alive, and treating those as one hobby with one health metric is the mistake. Count utility traffic, digital modes, pirate broadcasts, amateur-radio activity and numbers stations alongside international broadcasting, and the bands are, in the SWLing Post's own phrase, still "jammed" — just with different content than a decade ago, not with silence.[1] And a receiver that needs no internet connection, no cell signal and no functioning grid has genuinely provable value the one week those things fail — a case that has nothing to do with whether the sport of chasing weak, distant broadcast signals is as rich as it used to be.[1]

This book's own verdict, stated plainly rather than hedged into mush: the SPORT of classic broadcast DXing — chasing distant voices, verified by a mailed card — genuinely has less to chase than it did thirty years ago, and pretending otherwise would be dishonest. But the PRACTICE of shortwave listening, broadly defined to include decoding, monitoring and grid-down preparedness, has never had cheaper or better tools than it does right now, and the extended Solar Cycle 25 peak (p14) makes the next two to three years an unusually good window to be doing any version of it. Both halves of that sentence are true. A reader deciding whether to spend US$244 should weigh the version of the hobby he'd actually be practising, not the version Wikipedia is describing.

2 halves, 1 sentence

THE DECLINE CASE AND THE TOOLS-HAVE-NEVER-BEEN-BETTER CASE ARE BOTH GENUINELY TRUE — THEY'RE JUST ABOUT DIFFERENT VERSIONS OF THE SAME HOBBY[2]

The "jammed bands" claim has at least one hard number behind it that neither side of this argument usually cites: China alone counted roughly 240,000 licensed amateur-radio operators as of 2024, up from about 174,000 in 2021 — growth, not decline, in the one adjacent activity that shares the same spectrum, the same equipment culture, and often the same forums this book has mapped.[3] A shrinking broadcast band sitting inside a growing amateur-radio population is a genuinely odd picture, and it supports the "undead" thesis better than any single forum quote does.

174,000 → 240,000

CHINA'S LICENSED AMATEUR-RADIO OPERATORS, 2021 VS 2024 — REAL GROWTH IN THE ADJACENT ACTIVITY SHARING THIS HOBBY'S SPECTRUM AND CULTURE[3]

THE SCEPTIC'S LAST WORD

If, after the free month and the US$244 kit, this still feels like static and jargon rather than a genuine pull, that is a legitimate outcome this book will not talk a reader out of. A roughly two-hundred-and-fifty-dollar experiment is a genuinely cheap way to find out, against the far larger sums documented further up this book's own ladder — and Wikipedia's "few practical reasons" verdict, cited at the top of this section, will simply have been correct for that particular reader.[4]

WHERE THIS BOOK LANDS

Buy the US$244 kit expecting to practise the modern version of this hobby — decoding, monitoring, a grid-down safety margin — not the 1985 version of chasing Radio Moscow's English service. If that modern version still doesn't hold your attention after ninety days (p23), that is the same legitimate, cheap-to-discover outcome named above.[5]

SOURCES 1 The "undead" community rebuttal — swling.com/blog and London Shortwave blog commentary, as characterised in prior search-summarised community sentiment (single-sourced framing, flagged); cross-referenced against MERIDIAN No. 71 (daily), p17. 2 Editorial synthesis, this desk. 3 China amateur-radio operator count, ~240,000 (2024) vs ~174,000 (2021) — qsl.net/4x4xm, "Amateur Radio Operators by Country"; sarts.org.sg, "Introduction to Amateur Radio in China" PDF. 4 Wikipedia "few practical reasons" framing — see p19 sourcing. 5 See p23.

THE LOCAL ANGLE

The Little Red Dial

Singapore's part in this story starts in 1924 with the same amateur-wireless enthusiasm this whole book has been describing — and the one legal question every local reader actually wants answered turned out to be harder to pin down than this book expected.

this story👍👎

Singapore's radio history begins earlier and stranger than most residents realise. Broadcasting on the island started in 1924, and its earliest driving force wasn't a commercial company but a small circle of amateur radio enthusiasts organised as the Amateur Wireless Society of Malaya, sending shortwave signals from what was then a British colonial outpost.[1] The medium's wartime chapter is darker and directly continuous with this book's Long Wave section: after the British surrender, the Japanese occupying authorities seized the station and renamed it Syonan Hoso Kyoku — the "Light of the South" Broadcasting Corporation — resuming transmission on 13 March 1942 across three frequencies simultaneously: medium wave for the island itself, a relay of Tokyo programming, and a shortwave frequency aimed specifically at Australia.[1] Japanese-language lessons, mass-exercise programming, and instruction in showing proper respect for Japanese symbols filled the schedule; radio was, in the occupying administration's own use of it, the chief instrument for teaching the island a new imposed culture, not merely reporting news.[1] Visitors curious about the era's communications infrastructure more broadly can still walk through the Battlebox, the wartime underground command bunker beneath Fort Canning, whose more than twenty rooms include a dedicated signal and cipher control area — not radio-specific, but a genuine, visitable piece of the same period.[2]

The practical question every local reader actually wants answered is simpler and, honestly, harder to nail down than this book expected: is receive-only shortwave listening legal here without a licence? The daily flagged this as understood-but-unverified and asked readers to check before relying on it; this book tried to verify it properly, and reports the result honestly rather than upgrading an unconfirmed claim into a fact.[3] IMDA's published Telecommunications Act regulations and licensing guides, searched directly for this book, describe network, broadcast and equipment-type-approval licensing frameworks in detail but did not surface an explicit clause naming shortwave RECEIVE-only listening as licence-exempt. The community's working understanding — echoed by the daily and consistent with how virtually every other jurisdiction treats reception versus transmission — is that Singapore's licensing regime, like most, targets the OPERATION of a transmitter, not the act of listening. This book could not locate the specific statutory line that says so in plain terms, and says so plainly rather than pretending otherwise.[3]

1924

YEAR RADIO BROADCASTING BEGAN IN SINGAPORE — DRIVEN BY AMATEUR WIRELESS ENTHUSIASTS, NOT A COMMERCIAL BROADCASTER[1]

This book tried to verify the licensing claim harder than a Tuesday allows — and reports honestly where that verification ran out of road.

SOURCES 1 Singapore radio origins, Amateur Wireless Society of Malaya, 1924 — nlb.gov.sg, "Radio broadcasting in Singapore (1924–46)"; Syonan Hoso Kyoku, 13 Mar 1942, three-frequency structure — roots.gov.sg, "Celebrating Radio: Stories from the Past"; culturepaedia.singaporeccc.org.sg. 2 Battlebox, Fort Canning — battlebox.sg; Wikipedia, "Battlebox." 3 IMDA licensing framework search, this session — imda.gov.sg regulatory pages and iris.imda.gov.sg licensing guides (no explicit receive-only SWL clause surfaced); cross-referenced against MERIDIAN No. 71 (daily), p17, which carried the same unverified flag.

THE PRACTICAL REALITY

No Local Exam, One Electronics District, and a Dense Noise Floor

For a Singapore reader who wants to go further than listening and eventually transmit, the path is currently indirect. The Singapore Amateur Radio Transmitting Society confirms the local Radio Amateur Examination is presently suspended, with no local sitting available; SARTS instead recommends the FCC route — sitting a US Technician, General or Extra class exam — since IMDA recognises a valid FCC licence as the basis for issuing a Singapore amateur licence.[1] That means a Singapore SWL who eventually wants to transmit is, for now, studying American exam material and converting the result locally rather than sitting a Singapore-specific test — a genuinely unusual arrangement this book has not seen replicated in any other hobby it has covered.

On retail, this book found no dedicated shortwave or SWL-specific shop anywhere on the island; the closest thing to a walk-in radio-gear district is Sim Lim Tower on Jalan Besar, which houses at least two amateur-radio-adjacent dealers, Blazer Electronic Centre and Tecomart, alongside its general electronics-component trade — genuinely closer to a ham-radio and components resource than a shortwave-specific one, and most gear named across this book's Ladder and Ceiling sections still arrives here by international shipping from Anon-Co, GigaParts or a manufacturer's own shop.[2] This book also found no dedicated local SWL club, Telegram group or Facebook community during this session's search — worth stating plainly rather than inventing one for colour; whatever local scene exists appears to ride on SARTS' amateur-radio community and the international forums and Discords mapped on p7–8, not a standalone Singapore SWL institution.[3]

The daily already put a number on the FCC route's local cost — roughly S$50 a year through SARTS once the American exam is passed — and this book takes that figure as read rather than re-verifying it from scratch, per the continuity rule this magazine applies to its own prior reporting.[1] Compared with the SG$0 cost of simply listening, it is a meaningful but not prohibitive step up for a reader who eventually wants to transmit rather than only receive.

2

AMATEUR-RADIO-ADJACENT DEALERS CONFIRMED AT SIM LIM TOWER, SINGAPORE'S CLOSEST THING TO A RADIO-GEAR DISTRICT[2]

THE HDB REALITY — PRACTICAL ADVICE

The problem. A dense HDB block sits close to a worst-case radio-frequency noise environment, per the daily's own reporting and cross-checked against general RF-interference research on switching electronics in cities.[4]

The fix. A magnetic loop antenna (p15) placed at a window or balcony, fed with a common-mode choke to keep building-generated noise off the coax, is the single highest-leverage purchase for exactly this environment.[4]

WHAT ACTUALLY SHIPS HERE WITHOUT A FUSS

NASWA and WRTH membership. Both accept international subscribers by default (p8) — a Singapore reader loses nothing by joining either on day one.

Anon-Co and GigaParts. Both ship the Tecsun and Airspy lines internationally; budget for import shipping and any GST on arrival, same as any hobby gear ordered from overseas.

THE SCEPTIC, LOCAL — AN HONEST LIMIT ON THIS SECTION

This book leaned on the daily's prior HDB interference reporting rather than independently re-verifying a specific forum thread this session, and could not confirm an explicit IMDA receive-only exemption clause despite trying. Two honest holds in one section is more than most of this book carries — a fair reflection of how much less has been written about this hobby's Singapore specifics than about, say, its gear market.[5]

SOURCES 1 SARTS on RAE suspension and FCC route — sarts.org.sg, "Radio Amateur Examination (RAE)" and "FCC amateur radio licensing guide for Singapore HAMs"; ndoo.sg, "Obtaining an FCC Radio Amateur Technician License (for Singaporeans)." 2 Sim Lim Tower dealers — g4zfe.com, "radio Shopping in Singapore," listing Blazer Electronic Centre and Tecomart at 10 Jalan Besar. 3 Local-community search, this session — no dedicated Singapore SWL club, Telegram or Facebook group surfaced; held honestly rather than invented. 4 HDB noise-floor reality — MERIDIAN No. 71 (daily), p17; general urban RFI trend per RSGB, "The Background Noise on the HF Amateur Bands." 5 Editorial honesty note, this desk.

THE STARTING PATH

The First Year

Lurk before you spend, buy once when you do, and give yourself a full solar-cycle-favoured year before deciding which version of this hobby actually fits you.

this story👍👎

Month one costs nothing. Read the SWLing Post daily, subscribe to r/RTLSDR and r/AmateurRadio, and try a free WebSDR or KiwiSDR remote receiver through a browser to get a feel for tuning and the waterfall display before spending a dollar — an honest, if incomplete, taste test that p19's sceptic section takes seriously as "not quite the real thing," but a genuinely useful zero-cost one nonetheless.[1] Download free SDR software — SD++ or CubicSDR are the community's preferred starting points (p15) — and get comfortable reading a spectrum before owning hardware to point it at.

Weeks four through eight are for buying the real kit. This book's recommendation, matching the daily's own priced starter bundle, is roughly US$242: an RTL-SDR V4L (US$37.95) to keep learning the software with real hardware, an Airspy HF+ Discovery (US$169) as the receiver that actually rewards the investment, and an Airspy YouLoop (US$35) to beat whatever urban noise floor you're working against.[2] A reader who already knows they want a self-contained portable rather than a computer-tethered SDR can substitute a Tecsun PL-330 (~US$75–90) alone, upgradeable to a PL-880 (~US$165) later without wasting the first purchase.

Month 1

COST: US$0. LURK, TRY A FREE WEBSDR, AND GET COMFORTABLE WITH A WATERFALL DISPLAY BEFORE BUYING ANYTHING[1]

> ***Lurk before you spend, buy once when you do, and give yourself a full year before deciding which version of this hobby fits you.***

MONTHS THREE TO SIX — BUILD THE HABIT

Join NASWA's e-Journal (US$16/yr) for the weekly Flashsheet DX tip sheet — a curated, expert-run shortcut to what's actually worth chasing.[3]

Start a real log — paper or spreadsheet, date/time/frequency/station — the discipline that turns listening into DXing.

Catch the anchors. Radio Romania International, WWV time signals (2.5/5/10/15/20 MHz), and — if numbers stations intrigue you — UVB-76 on 4625 kHz via Priyom's published schedules.[4]

THE COMMUNITY'S OWN ADVICE TO BEGINNERS

Asked what a first-year listener most often gets wrong, the recurring answer across the SWLing Post and r/RTLSDR is the same one this book has repeated throughout: buying a second receiver before buying a real antenna. The community's consistent advice is to spend the first six months mastering ONE receiver and ONE loop antenna before adding anything else — the instinct to diversify gear early is, by the community's own account, the single most common way a beginner's budget disappears without a matching improvement in what they can actually hear.[5]

THE SCEPTIC, EARLY — DON'T OVER-PLAN YEAR ONE

This whole page is a plan, and plans are exactly what a hobby like this one tends to defeat. The honest caveat: buy the US$244 kit, then let curiosity — not this schedule — decide what happens in month four. A reader who's still only using the free WebSDR by month six hasn't failed the programme; the programme was always optional scaffolding, not a test.[6]

SOURCES 1 WebSDR/KiwiSDR as a free entry point — see p7 sourcing. 2 Starter-kit pricing — see p9, p17 sourcing. 3 NASWA — see p8 sourcing. 4 Radio Romania International, WWV — MERIDIAN No. 71 (daily), p16; UVB-76/Priyom — see p6 sourcing. 5 Community beginner-advice pattern — characterised from swling.com/blog and r/RTLSDR forum-tone research across this book, this desk. 6 Editorial framing, this desk.

MONTHS SIX TO TWELVE

Pick Your Tribe — and a Real Shopping List

By month six, the honest signal to read is which version of this hobby is actually holding your attention, and to spend the second half of the year accordingly rather than buying everything at once. If distant, weak signals are the draw — the DXer's path — the next purchase is a proper boutique loop antenna (p12), from an MLA-30+ at US$30–37 up to a W6LVP at US$445–495, whichever the budget and the noise floor demand. If the technology and decoding side is the draw, the next purchase is an RX-888 MkII (~US$170) for wideband recording and a real dive into FT8, HFDL and WEFAX (p15). If grid-down resilience is the real motivation, the next purchase is a rugged, battery-friendly portable — the Tecsun PL-880 again, plus spare 18650 cells and a small solar charger — over anything computer-dependent.[1]

None of these paths is more legitimate than the others, and nothing about this book's own buy-once-cry-once framing requires picking only one forever. What it does argue for is picking deliberately, twelve months in, rather than accumulating gear reflexively the way the Sceptic section's "money pit" warnings describe (p19). A reader who reaches month twelve still logging catches, still checking the SWLing Post, and still enjoying the ritual has, by this hobby's own modest standards, arrived — no further purchase is required to belong.

US$242–$700

THE REALISTIC FULL FIRST-YEAR SPEND FOR A SERIOUS, BUY-ONCE BEGINNER, ACROSS EITHER THE DXER OR THE TECHNOLOGY PATH[2]

> *A reader who reaches month twelve still logging catches, still checking the SWLing Post, and still enjoying the ritual has arrived — no further purchase is required to belong.*

THE FIRST-YEAR SHOPPING LIST, PRICED

ITEM	PRICE
RTL-SDR Blog V4L	US$37.95
Airspy HF+ Discovery	US$169
Airspy YouLoop	US$35
NASWA e-Journal, 1 year	US$16
Optional: W6LVP loop upgrade	US$445–495

FOUR MILESTONES WORTH MARKING

The first real DX catch. A broadcaster from a continent away, logged with date, time and frequency — the moment the daily's own "gateway" description stops being theoretical.

The first decode. An FT8 signal, a WEFAX chart or an HFDL frame pulled cleanly out of noise you couldn't otherwise hear — the technology path's equivalent gateway moment.

The first numbers station. UVB-76's buzz, or any ENIGMA-listed station, logged and cross-checked against Priyom's schedules — the moment the hobby's strangest corner stops being a rumour.

The first upgrade decision made deliberately. Not an impulse buy, but a considered move up this book's own Ladder or Ceiling — the moment "buy-once-cry-once" stops being a slogan and starts being a lived habit.

> **WHERE THIS BOOK GOES NEXT**
>
> Everything above assumes shortwave stays roughly what it is today — a shrinking broadcast band riding inside a growing, well-tooled listening and decoding culture. The Long Read that closes this book (p25) tests that assumption against the one place this hobby's oldest use — carrying a signal past a government that doesn't want it heard — is still being fought over, in real time, as this book goes to press.[3]

SOURCES 1 Path-specific gear — see p9, p12, p13 sourcing. 2 First-year spend synthesis — this desk, cross-referenced against p17. 3 Editorial handoff, this desk.

THE LONG READ

The Keepaway Broadcast

Somewhere between Taiwan, Thailand and mainland China, more than a hundred antennas are playing a live game of keepaway with a state jammer that has never once identified itself on air — and Western press has mostly filed it as a footnote to a 2018 court case rather than the still-unfolding radio war it actually is.

this story👍👎

Every mechanism this book has spent twenty-four pages pricing and mapping — wideband capture, loop antennas, the discipline of logging a weak signal against noise — is exactly the toolkit someone would need to follow the strangest live story in shortwave today, and almost nobody covering it has read it that way. Sound of Hope is a Chinese-language radio network founded in 2004, part of a media constellation associated with the Falun Gong movement alongside outlets like Epoch Times and NTD, broadcasting uncensored news into mainland China via shortwave from transmitter sites in Pingtung and Miaoli, Taiwan.[1] Every transmitter runs at 1 kilowatt or less — a fraction of what the BBC World Service historically used — which should make the network trivial for a state broadcaster to jam off the air entirely.[1]

It hasn't been, because of a deliberately eccentric strategy the network's own operators and outside reporters describe as playing keepaway. Rather than broadcasting from one fixed site, Sound of Hope operates more than a hundred antennas spread across countries surrounding China — Thailand and Taiwan chief among them — trading the broadcast between sites on frequencies deliberately outside the bands a target broadcaster normally uses, making the signal harder for automated jamming to lock onto.[2] It is, functionally, the same multi-frequency anti-jamming logic Radio Free Europe used against Soviet jammers (p5) — run not by a state department's budget but by a diaspora religious movement's volunteers.

100+

ANTENNAS SOUND OF HOPE OPERATES ACROSS COUNTRIES SURROUNDING CHINA, DELIBERATELY SHUFFLED AND OUT-OF-BAND TO EVADE JAMMING[2]

> ***The same multi-frequency anti-jamming logic the Cold War's Radio Free Europe used against Soviet jammers — run not by a state department's budget but by a diaspora religious movement's volunteers.***

THE JAMMER THAT NEVER SPEAKS

China's answer runs through CNR-1, which deploys the "Firedrake" jammer against Sound of Hope, Radio Free Asia, Voice of Tibet and Radio Taiwan International alike — a loop of thirteen Chinese folk songs, relayed via ChinaSat 6B directly onto its targets' own frequencies. It has never once carried a spoken identification. Its only function is denial, dressed as music.[3]

TAIPEI — **SOUND OF HOPE'S OWN FRAMING**

A distributed, low-power network broadcasting uncensored news to a population that cannot otherwise receive it — the antenna-shuffling as defensive necessity (p5).[4]

BEIJING — **CNR-1'S IMPLICIT FRAMING**

An unlicensed foreign broadcaster tied to a banned religious movement — Beijing's general position on Falun Gong-linked media treats it as a hostile information operation, not a press-freedom question.[4]

SOURCES 1 Sound of Hope founding, 2004, Falun Gong-affiliated network, Pingtung/Miaoli Taiwan sites, ≤1kW power — grokipedia.com/page/Sound_of_Hope; rsf.org. 2 100+ antenna "keepaway" strategy — hard-core-dx.com DX schedules; defenseone.com, "How Dissidents Are Using Shortwave Radio to Broadcast News Into China" (2019); secondary/aggregated reporting, flagged. 3 Firedrake — sigidwiki.com; hfunderground.com wiki; CNR-1 targeting — Wikipedia, "Radio jamming in China." 4 Viewpoint framing — editorial synthesis of sources 1–3 and p5.

THE HUMAN COST, AND THE PRESENT TENSE

A Trial That Never Fully Resolved, and a Station Shut Down Last Year

The people running these antennas carry real legal risk. In August 2018, Thai officials shut down a Sound-of-Hope-linked shortwave station operating from a property in Chiang Mai; Yung-hsin Chiang, a 52-year-old Taiwanese businessman who had rented the property, was arrested that November and charged with operating a radio station without a permit, facing up to five years in prison. He denied direct involvement, saying the broadcasting equipment belonged to his tenants.[1] Reporters Without Borders publicly called on Thai authorities to drop the case, and a 2013 Freedom House report — cited by RSF in Chiang's defence — had already documented a pattern of Chinese diplomatic pressure on Thailand and neighbouring governments specifically to shut Sound of Hope stations down.[2] This book could not confirm the trial's final verdict through this session's research and holds that fact honestly rather than guessing — the most recent reporting located dates to the original 2019 hearing coverage.[1]

The pressure campaign is not history. A separate Sound-of-Hope-linked shortwave station that had operated openly from Chiang Mai since 2015 was shut down by Thai authorities in August 2024 — six years after Chiang's arrest, and well inside the twelve months this book's Cutting Edge section (p13–14) covers for every other part of this hobby.[3] Read against everything else in this book, the contrast is the actual argument. The Long Wave chapter (p5–6) told a story of broadcasters LEAVING shortwave voluntarily, priced out by satellite and streaming economics. This is the opposite pole: a broadcaster fighting, at real legal risk to the people renting it antenna space, to STAY on a band that a state is actively spending resources to deny it — using, stripped of the politics, exactly the multi-frequency, keep-moving anti-jamming logic this book traced back to the 1950s on p5.

Aug 2024

A SOUND-OF-HOPE-LINKED CHIANG MAI STATION, OPERATING SINCE 2015, SHUT DOWN UNDER CHINESE DIPLOMATIC PRESSURE — INSIDE THIS BOOK'S OWN TWELVE-MONTH "CUTTING EDGE" WINDOW[3]

A medium that is dying by neglect in the West is being actively contested by force in Asia — and both stories are true at the same time, about the same band.

A NOTE ON THIS STORY'S SOURCING

This account rests almost entirely on Taiwanese and Western secondary reporting — RFA, BenarNews, RSF, and search-aggregated summaries — rather than a direct statement from the Sound of Hope network itself, which this book could not independently reach. That is itself a small illustration of the story's own thesis: even a magazine built to hunt down under-covered Asia-Pacific stories found this one easier to trace through Western press-freedom coverage than through the network at its centre.[4]

WHAT THIS BOOK WOULD WATCH NEXT

Whether Lampertheim's VOA testing (p6) scales into a genuine broadcast schedule — or stalls as a symbolic gesture that never grows past a few minutes a day.

Whether Sound of Hope's antenna network absorbs the loss of another host country the way it evidently absorbed Chiang Mai's 2024 closure — or whether the keepaway strategy has a genuine breaking point this book's research didn't surface.

SOURCES 1 Chiang Yung-hsin arrest and charges — benarnews.org and rfa.org, "Taiwanese Man Faces Thai Trial for Allowing Radio Broadcasts into China," 13 Jun 2019; verdict not confirmed by this session's search, held honestly. 2 RSF statement and 2013 Freedom House report reference — ladailypost.com, "RSF: Prosecution Over Shortwave Broadcasts To China"; rsf.org. 3 August 2024 Chiang Mai station closure — search-synthesised current-affairs reporting, cross-checked against the established 2015 station-opening date in prior coverage; single-sourced on the exact 2024 closure date, flagged. 4 Editorial sourcing note, this desk.

THE LAST WORD

A dial with nothing on it is still a dial worth owning — that was this book's opening paradox, and twenty-four pages later it holds up better than expected. The broadcasters really are leaving, the noise floor really is worse, and the QSL card really has mostly disappeared. But the tools have never been cheaper, the propagation has rarely been better, and somewhere between Taipei and Bangkok a hundred antennas are proving, on live evidence, that a government still spends real money trying to silence a band it could simply have ignored if the band no longer mattered.

FULL SOURCE LIST

MERIDIAN No. 71 (daily), "The Rabbit Hole · The Band Everyone Left" · SWLing Post (swling.com) · QRPer.com · Shortwave Radio Audio Archive · r/AmateurRadio · r/RTLSDR · r/shortwave · SubredditStats · RedditList · HFUnderground (hfunderground.com) · Priyom.org · ENIGMA 2000 (signalshed.com) · NASWA (naswa.net) · World Radio TV Handbook / Radio Data Center (wrth.info, radiodatacenter.net) · Wikipedia (Shortwave listening, UVB-76, R-390A, Radio jamming in China, Battlebox, Radio Netherlands Worldwide) · Cold War Radio Museum · Radio World · Al Jazeera · NPR · KJZZ · nerfd.net · rtl-sdr.com · airspy.com · sdrplay.com · rx-888.com · elekitsorparts.com · wsjtx.github.io · spaceweather.gov (NOAA SWPC) · moonrakeronline.com · anon-co.com · gigaparts.com · universal-radio.com · dxengineering.com · radioblvd.com · 1919a4.com forums · eham.net · collinsradio.org · w6lvp.com · bonito.net · swling.com/blog "Living in a World without Wellbrook" (sdxf.se) · tindie.com/SofterHardware · GitHub, softerhardware/Hermes-Lite · radiotapes.com · english.radio.cz · portableoperator.com · DataIntelo · 360iResearch · Cognitive Market Research · adventure-wiser.com · survivalistprepper.net · IMDA (imda.gov.sg, iris.imda.gov.sg) · SARTS (sarts.org.sg) · ndoo.sg · g4zfe.com · nlb.gov.sg · roots.gov.sg · Culturepaedia Singapore Chinese Culture Centre · battlebox.sg · Grokipedia · RSF (rsf.org) · Radio Free Asia · BenarNews · Defense One · sigidwiki.com · hard-core-dx.com.